

Deriving p^2 from x , y , and z

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User prompt: given $z = a / w / \sqrt{p^2 - a^2}$ $y = p^2 / w / (p^2 - a^2)^{3/2}$ $x = 1 / w / \sqrt{p^2 - a^2}$ find a compact expression for p^2 from x , y , and z

Response: To find a compact expression for p^2 in terms of x , y , and z , we can manipulate the given equations algebraically.

1. Analyze the equations:

The given equations are:

$$1. \quad z = \frac{a}{w \sqrt{p^2 - a^2}}$$

$$2. \quad y = \frac{p^2}{w(p^2 - a^2)^{3/2}}$$

$$3. \quad x = \frac{1}{w \sqrt{p^2 - a^2}}$$

2. Express a in terms of z and x :

Divide equation (1) by equation (3):

$$\frac{z}{x} = \frac{\frac{a}{w \sqrt{p^2 - a^2}}}{\frac{1}{w \sqrt{p^2 - a^2}}} = a$$

Thus, we have:

$$a = \frac{z}{x}$$

3. Express the ratio y/x :

Divide equation (2) by equation (3):

$$\frac{y}{x} = \frac{\frac{p^2}{w(p^2 - a^2)^{3/2}}}{\frac{1}{w \sqrt{p^2 - a^2}}}$$

$$\frac{y}{x} = \frac{p^2}{w(p^2 - a^2)^{3/2}} \cdot w \quad p^2 - a^2$$

$$\frac{y}{x} = \frac{p^2}{p^2 - a^2}$$

4. Solve for p^2 :

Let $R = \frac{y}{x}$. The equation becomes:

$$R = \frac{p^2}{p^2 - a^2}$$

Rearrange to solve for p^2 :

$$R(p^2 - a^2) = p^2$$

$$Rp^2 - Ra^2 = p^2$$

$$Rp^2 - p^2 = Ra^2$$

$$p^2(R - 1) = Ra^2$$

$$p^2 = \frac{Ra^2}{R - 1}$$

5. Substitute R and a back into the expression:

Substitute $R = \frac{y}{x}$ and $a = \frac{z}{x}$:

$$p^2 = \frac{\left(\frac{y}{x}\right) \left(\frac{z}{x}\right)^2}{\frac{y}{x} - 1}$$

Multiply the numerator and the denominator by x to simplify:

$$p^2 = \frac{y \left(\frac{z}{x}\right)^2}{y - x}$$

$$p^2 = \frac{yz^2}{x^2(y - x)}$$

Final Compact Expression:

$$p^2 = \frac{yz^2}{x^2(y-x)}$$
